

In any organization—be it industrial, commercial, or corporate—accurate employee attendance and work hour tracking are essential components of effective workforce management. Most companies still rely on traditional methods such as manual registers, biometric systems, or RFID-based attendance systems. These methods, while common, come with several limitations such as human error, buddy punching, limited accuracy, maintenance issues, and a lack of real-time monitoring capabilities.

With the increasing demand for automation and the rapid evolution of Internet of Things (IoT) technologies, it has become necessary to rethink and upgrade the conventional methods of attendance management. Our project titled “**BLE-Based Employee Tracking System**” aims to bridge this technological gap by offering a smart, cost-effective, and scalable solution using Bluetooth Low Energy (BLE) beacons and ESP32 microcontrollers.

The system is designed in such a way that each employee carries a small BLE tag, specifically using the SparkFun NanoBeacon IN100. These tags constantly emit a unique signal (UUID), which can be detected by an ESP32-based BLE receiver placed at the main entrance of the workplace or any designated checkpoint. The ESP32 captures the UUID, logs the exact entry or exit time, and forwards this data to a cloud backend system (e.g., Firebase) via Wi-Fi connectivity.

This data is then processed and displayed on a user-friendly web dashboard that can be accessed by administrators or managers in real time. The system calculates the total active work hours per employee on a daily basis, which helps in maintaining accurate attendance records and streamlining payroll calculations.

Unlike traditional methods, this system provides multiple benefits including reduced human error, improved transparency, increased data accuracy, and reduced administrative workload. It also eliminates the need for physical touch (as required in biometric systems), which makes it more hygienic—especially relevant in a post-pandemic world.

Furthermore, the system’s low-power design ensures that the BLE beacons can last for several months to a year on a single coin cell battery, making it both sustainable and easy to maintain. The ESP32-based receiver is programmable, scalable, and cost-efficient, making it suitable for small, medium, or large enterprises.

In conclusion, the **BLE-Based Employee Tracking System** is an innovative step toward automating workforce management using modern IoT solutions. It reflects the growing trend of smart infrastructure and promises to improve overall operational efficiency, reduce costs, and provide managers with reliable data for better decision-making.